

Memorandum

To leadership

From: Kevin Kallish

Date: 04/26/25

Subject: Analysis of the National Cybersecurity Strategy's Pillar on Defending Critical Infrastructure

Purpose

This memo provides an analytical overview of the 2023 White House National Cybersecurity Strategy. It focuses on the U.S. government's recommendations for strengthening the defense of critical infrastructure that supports the development, design, production, delivery, and sustainment of military weapon systems, subsystems, and components vital to U.S. battlefield readiness.

Our organization's security posture directly contributes to helping maintain the U.S. military's technological edge over adversaries. The technology we develop and the data we manage are valuable targets. If our systems are compromised, they could be weaponized against the United States.

Background

Espionage from China, Russia, and North Korea has reached unprecedented levels. Today, nearly 4,000 cyberattacks occur daily, and every 14 seconds, a new company falls victim to a ransomware attack. According to Astra Security, over 560,000 new types of malware are detected daily (Astra Security, 2024)¹. Our organization's work is highly strategic, making us a target of espionage by U.S. adversaries. Foreign adversaries are probing our networks around the clock searching for vulnerabilities they can exploit to access sensitive American defense technology. The main focus areas of foreign intelligence agencies are stealth technology, submarine capabilities, missile guidance systems, and applications of AI and semiconductors. China, Russia, and North Korea seek to understand, reverse engineer, and replicate our systems to accelerate their military capabilities, allowing them to bypass years of costly research and development.

Analysis

The White House released a 2023 National Cybersecurity Strategy outlining five key pillars for defending the nation's critical infrastructure. At the forefront is the need to strengthen partnerships between the U.S. government and the private sector. As the report highlights, most of America's critical infrastructure is owned and operated by private entities who do not share cyber intelligence. The federal government is pushing to boost sharing cyber threat intelligence between private businesses to protect our national security. Agencies such as CISA will take the

¹ Astra Security. (2024). *How many cyber attacks happen per day?* Retrieved from <https://www.getastra.com/blog/security-audit/how-many-cyber-attacks-per-day/>

lead in ensuring that threat information is shared quickly across sectors and in coordinating responses when incidents occur. The strategy clarifies that active information sharing and working together on defense plans isn't optional. With adversaries constantly targeting these systems to gain access to sensitive and strategically essential technologies, collaboration is more critical than ever.

The report signals a clear shift from voluntary compliance to enforceable cybersecurity requirements across all critical infrastructure sectors. Each sector will be overseen by its regulatory agency, which will begin developing and enforcing minimum cybersecurity standards. For the defense industrial base, this means complying with standards set by the Department of Defense. At the same time, other sectors will fall under the jurisdiction of agencies such as the Department of Energy, Transportation Security Administration, Environmental Protection Agency, and others. These standards will create clear expectations across industries for managing risk, maintaining continuity, and building resilience so organizations are better prepared to prevent and bounce back from cyberattacks. Organizations that lack strong security policies and procedures don't just face potential risks- they can incur serious financial consequences. Most importantly, the strategy treats cybersecurity as a public safety imperative: a breach in any critical sector could have cascading and severe consequences for the American public.

While the private sector will be held to strict compliance standards by its respective regulatory agencies, federal agencies must adopt modern, secure-by-design cybersecurity practices. A central pillar of this initiative is the implementation of Zero Trust Architecture, which applies rigorous access controls throughout the organization. This model operates on the principle that no user or system is inherently trusted—every access request, regardless of its origin, must be thoroughly authenticated and authorized before any access is permitted. Users will operate with the least privilege access, meaning they are only given the minimum access necessary to perform their duties.

Endpoint security is a top priority, as endpoints are treated as untrusted by default until fully authenticated. The federal government is also pushing to modernize its threat detection, automated incident response, and cross-agency coordination capabilities. These efforts are intended to set a standard where federal leadership in cybersecurity can help reduce the overall attack surface and strengthen resilience across both the public and private sectors.

Cyber defense requires unified, real-time coordination across all government and industry sectors. The report underscores the importance of CISA, particularly the Joint Cyber Defense Collaborative (JCDC), as a key driver in unifying cybersecurity efforts across the nation's critical infrastructure. The JCDC is responsible for advancing joint planning, enabling proactive threat detection, and leading coordinated cyber exercises across sectors to strengthen collective defense. The central government body drives faster, more integrated responses to significant cyber incidents. By fostering operational collaboration—such as red team/blue team simulations—the JCDC enables organizations to test and strengthen their cyber defenses in a unified, strategic manner.

Modern infrastructure depends on a mix of cloud platforms and older industrial systems—both of which can be easy targets if not properly secured. The report stresses the need for secure-by-default settings in cloud environments, like encryption, activity logging, and

multi-factor authentication. For organizations using cloud—especially hybrid or multi-cloud setups—it’s critical to have complete visibility and control over those environments. Shared responsibility models between cloud providers and their customers are also essential to avoid gaps in security and accountability.

Regarding operational technology (OT), network segmentation is key in stopping threat actors from moving laterally within systems. Legacy systems that control physical processes must be regularly patched and updated, and a wider effort should be made to modernize this infrastructure. Strengthening supply chain security and oversight is essential to reduce broader vulnerabilities across connected systems.

Recommendations

To ensure compliance with the requirements outlined in the U.S. Government’s National Cybersecurity Strategy (2023), we must implement the following across our organization:

1. Conduct Proactive Gap Assessments

Routinely evaluate our cybersecurity posture against key frameworks, including CMMC 2.0 Expert, NIST SP 800-171, NIST SP 800-172, NISPOM, and NIST SP 800-37. These proactive assessments help us identify and remediate compliance gaps before they jeopardize our eligibility for Department of Defense (DoD) contracts.

2. Establish a Cyber Liaison Team

Designate a dedicated cyber liaison team to actively engage with Information Sharing and Analysis Centers (ISACs) and the Joint Cyber Defense Collaborative (JCDC). These liaisons will facilitate real-time, bidirectional threat intelligence sharing, improving situational awareness across public and private sector threat landscapes.

3. Evaluate and Strengthen Cloud Security

Review and harden our cloud environments by enforcing secure-by-default configurations such as encryption, access logging, and robust identity and access management (IAM) controls. A clear definition of roles and responsibilities under the shared responsibility model with cloud service providers will further ensure compliance and reduce risk.

4. Run Regular Cybersecurity Exercises

Conduct regular tabletop exercises and red/blue team simulations in coordination with internal stakeholders and public sector partners. These exercises, aligned with DoD cybersecurity frameworks, will test our incident response plans against realistic and evolving threat scenarios.

5. Prioritize Workforce Cybersecurity Training

Recognizing that our people are the first line of defense, we will implement ongoing cybersecurity training programs. These will include mandatory phishing simulations, baseline

security education for all employees, and advanced technical training for specialized teams to mitigate human-related vulnerabilities.

6. Adopt a Zero Trust Security Model

Implement a Zero Trust Architecture (ZTA) that includes micro-segmentation, least privilege access, and continuous validation of users, devices, and applications. This model ensures that no internal or external actor is trusted by default, significantly enhancing our defense posture.

By implementing these recommendations as soon as possible, our organization will meet federal compliance standards while establishing itself as a leader in cybersecurity within the defense sector—clearly signaling our commitment to protecting sensitive, high-value data.